Project Design Phase
Proposed Solution Template

Date	03 April 2026
Team ID	SWTID-2026-8629
Project Name	Smart Library
Maximum Marks	2 Marks

Model Performance Testing – Smart Library System

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1. Data Rendered

The dataset containing library details such as Book Name, Author, Category, Availability, Issue Status, and User details was successfully imported and rendered in the dashboard. Data is displayed clearly using tables and visual components.

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2. Data Preprocessing

Data preprocessing was performed to clean and transform the dataset. Missing values were handled, duplicate book records were removed, and columns were formatted properly (e.g., issue/return dates, numeric values for fines). This ensures accurate analysis and visualization.

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3. Utilization of Data Filters

Filters and slicers were used to allow users to interact with the data. Users can filter information based on:

- **Author**
- **Book Category**
- **Availability status**

- **Issue/Return date**

This improves usability and helps in focused analysis.

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4. DAX Queries Used

DAX (Data Analysis Expressions) was used to create calculated fields and measures such as:

- **Total Books**
- **Available Books**
- **Issued Books**
- **Overdue Books**
- **Fine Amount**

These calculations enhance the analytical capability of the dashboard.

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5. Dashboard Design

The dashboard is designed to be user-friendly and visually appealing. It includes multiple charts such as bar graphs, pie charts, and line charts to represent library data effectively.

- **Number of Visualizations / Graphs: 6**

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6. Report Design

The report is structured into sections for better understanding, including:

- **Library overview**
- **Book availability analysis**
- **Issue and return trends**

Visual elements are arranged neatly for easy navigation.

- **Number of Visualizations / Graphs: 5**

Conclusion:

The dashboard and reports effectively present library data with proper visualization and interactivity. The use of preprocessing, filters, and DAX queries ensures accurate and meaningful insights, improving overall library management efficiency.